

10. (a) The table shows information about some Group 1 elements.

Element	Relative atomic mass	Number of electrons in the outer shell	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)
lithium	7	1	180	1342	0.53
sodium	23	1	98	883	0.97
potassium	39	1	63	759	0.89
rubidium	85	1	39	688	1.53
caesium	134	1	29	671	1.93

Use the information in the table to answer parts (i) and (ii).

- (i) State the information which explains why the elements have similar chemical properties.

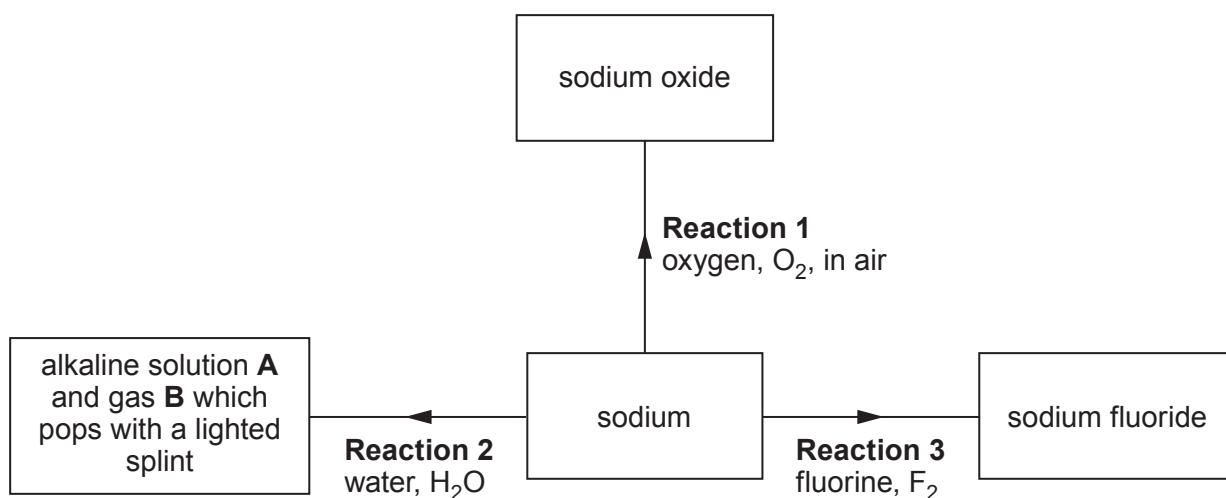
[1]

- (ii) State which **property** has a value which does **not** fit the trend down the group. [1]

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(b) The flow diagram shows some reactions of sodium.



(i) State how **Reaction 1** is prevented when storing sodium in the laboratory. [1]

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(ii) Give the names of alkaline solution **A** and gas **B**. [2]

..... and

(iii) Name the Group 1 metal which would react **least** violently with water. [1]

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(iv) Complete the symbol equation for **Reaction 3**. [1]



- (c) Sodium fluoride is added to some UK public water supplies to reduce tooth decay in children.

In America sodium hexafluorosilicate, Na_2SiF_6 , is more commonly used. The relative formula mass of sodium hexafluorosilicate is 188.

- (i) Calculate the percentage of fluorine in sodium hexafluorosilicate. [2]

$$A_r(\text{F}) = 19 \qquad M_r(\text{Na}_2\text{SiF}_6) = 188$$

Percentage = %

- (ii) State an **ethical** reason why some people oppose the fluoridation of water supplies. [1]

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- (iii) Apart from water supplies, state the most commonly used source of fluoride to reduce tooth decay. [1]

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END OF PAPER

